

# Detecting Misinterpreted but Genuine Citations Through Automated Claim Verification

## Abstract

Modern scientific literature relies heavily on bibliographic citations to anchor claims, trace intellectual lineages, and establish foundational validity. However, an insidious failure mode in academic writing is the proliferation of misinterpreted genuine citations—instances where an existing, legitimate publication is cited, but the referencing text distorts, overstates, or contradicts the original source's findings. Unlike hallucinated references generated by generative language models or blatant bibliographic errors, misinterpreted genuine citations pass superficial automated verification (such as Digital Object Identifier (DOI) resolution) because the cited document authentically exists. Detecting these semantic misalignments requires sophisticated Automated Claim Verification (ACV) architectures capable of cross-document natural language inference (NLI), citation context extraction, and deep contextual reasoning. This paper formalizes the problem of misinterpreted genuine citation detection, maps the pipeline of context-aware ACV frameworks, evaluates the state of deep learning models (including fine-tuned transformer architectures and multi-agent systems), details key technical challenges such as scope drift and full-text retrieval limitations, and outlines critical research gaps and future directions for semantic citation auditing in scientific communication.

**Keywords:** Automated Claim Verification, Citation Misinterpretation, Natural Language Inference, Quotation Errors, Scholarly Text Processing, Epistemic Integrity, NLP.

## 1. Introduction

Scientific progress is fundamentally cumulative. Researchers build upon prior findings, referencing foundational works to justify hypotheses, contextualize empirical data, and validate theoretical paradigms. To maintain this architecture of knowledge, the validity of scientific citations is paramount. Traditionally, citation integrity was evaluated primarily through bibliographic metadata accuracy—verifying whether an author, publication title, journal name, and DOI correctly matched an existing record in indexes like PubMed or Scopus.

However, a structural vulnerability in scholarly communication lies in semantic citation fidelity: whether the assertion made in the citing text (citing claim) accurately reflects the empirical evidence, conclusions, or nuances of the referenced source (cited evidence). Research suggests that citation and quotation errors are pervasive in academic literature; for instance, a significant percentage of references in scientific papers suffer from errors, often taking the form of factual inaccuracies or unsupported claims. This phenomenon—the **misinterpreted genuine citation**—occurs when a cited paper is authentic and correctly indexed, yet the citing document attributes findings to it that are exaggerated, taken out of context, selectively misquoted, or explicitly contradicted by the original text.

WJMH :: World Journal of Men's Health+ 1

Unlike "hallucinated citations" manufactured by Large Language Models (LLMs), misinterpreted genuine citations bypass basic automated checks. A DOI link resolves cleanly to a published paper, but the scientific claim attached to it is semantically false. When

uncorrected, misinterpreted citations create "information cascades," where a single secondary distortion is uncritically cited by subsequent authors, transforming an unsubstantiated claim into accepted domain consensus.

Automated Claim Verification (ACV) systems designed for the scientific domain offer a promising technological countermeasure. However, adapting ACV to distinguish subtle semantic distortions in authentic citations presents unique computational and linguistic challenges. This paper investigates automated detection of misinterpreted genuine citations, analyzing modern natural language processing (NLP) pipelines, evaluating recent multi-agent architectures, identifying technical bottlenecks, and highlighting avenues for future research.

## **2. Background and Related Work / Literature Review**

### **2.1 The Concept of Citation Distortion**

The sociological and epistemic consequences of citation misinterpretation have been studied extensively. Greenberg (2009) demonstrated how "citation distortions" (including citation bias, amplification, and invention) can create unfounded authority. When a paper claims something that its cited source does not substantiate, and this error is propagated, it establishes a false factual lineage. In biomedical domains, the prevalence of such quotation errors has historically hovered around 20%, often heavily distorting the original scientific evidence.

Semantic Scholar+ 1

Agarwal et al. (2023) cataloged citation errors into distinct categories, including "factual errors" (incorrect description of findings) and "selective citation" (ignoring

suitable context). These semantic errors are profoundly different from simple bibliographic mistakes, as they actively warp the scientific record. [PMC - NIH](#)

## 2.2 Evolution of Automated Claim Verification (ACV)

Early ACV frameworks focused primarily on open-domain fact-checking against unstructured web sources or Wikipedia data. The introduction of the FEVER dataset established a paradigm for fact extraction and verification utilizing machine learning.

Applying this paradigm to scientific literature introduced unique complexity. Scientific claims rely on dense domain terminology, precise quantitative bounds, and highly nuanced conditional logic. The introduction of the **SciFact** dataset formally bridged this gap by pairing expert-authored scientific claims with evidence sentences extracted from biomedical abstracts. In SciFact, systems must predict whether a source abstract [SUPPORTS](#) or [CONTRADICTS](#) a claim, or provides [NOT ENOUGH INFO](#). More recent corpora, such as those introduced by Sarol et al. (2024), have further refined this by focusing explicitly on in-text quotation errors and citation context integrity within full biomedical publications. [Kaggle+ 1](#)

## 3. Main Thematic or Technical Discussion

Detecting whether a genuine citation has been misinterpreted requires moving beyond single-document context toward cross-document semantic matching. The computational pipeline generally comprises four primary stages:

### 3.1 Citation Context Extraction

The system must first isolate the exact textual span where a claim is

made. This involves identifying the citation marker (e.g., [12] or (Smith et al., 2021)) and extracting the local sentence boundaries. Because single sentences often lack necessary anaphoric references (e.g., "They demonstrated that..."), modern models frame this as a sentence-level binary classification task. For instance, PubMedBERT models have been fine-tuned to predict whether surrounding sentences in a paragraph organically belong to the specific citation context. PMC - NIH

### 3.2 Source Retrieval and Alignment

The cited identifier (DOI, PubMed ID) is resolved against structured repositories to pull the target document. Unlike standard open-domain claim verification, the target document is known a priori. The challenge transitions from global document retrieval to fine-grained local evidence localization.

### 3.3 Fine-Grained Evidence Sentence Selection

Because full scientific papers contain hundreds of sentences, feeding the entire document into an entailment model introduces noise and exceeds context length limits. Evidence sentence retrieval is typically treated as a text-pair classification or ranking problem.

PMC - NIH

State-of-the-art methods leverage dual-encoder retrieval frameworks (e.g., BM25) combined with neural rerankers. For instance, a BM25 model can retrieve the top 60 candidate sentences from the reference text, which are then passed to a MonoT5 reranker to identify the top  $k$  sentences most semantically relevant to the citing claim.

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### 3.4 Semantic Natural Language Inference (NLI)

**Verification** The candidate evidence sentences and the citing claim are processed by an NLI engine to predict an epistemic relation label, often classifying the citation as

`ACCURATE`, `NOT_ACCURATE` (contradiction/distortion), or `IRRELEVANT`. This step hinges on the model's ability to detect whether the linguistic modality and empirical scope of the cited evidence match the citing claim.

ResearchGate

## 4. Current Approaches and Developments

### 4.1 Transformer-Based Verification Pipelines

Recent implementations have demonstrated moderate success using supervised deep learning. Sarol et al. (2024) introduced a pipeline utilizing a fine-tuned MultiVerS model for citation accuracy classification. Their approach—using BM25 for retrieval, MonoT5 for reranking, and a specialized classification head—achieved a macro-F1 score of 0.52 in detecting citation misalignments. While this proves the feasibility of automated auditing, it also highlights how subtle "quotation errors" remain computationally challenging for standard transformers.

PubMed+ 1

### 4.2 Large Language Models and Interpretability

With the advent of LLMs, few-shot in-context learning has been applied to quotation error detection. While base LLMs like GPT-4 perform decently well on `ACCURATE` classifications, they often lag in correctly identifying subtle erroneous citations without specific fine-tuning. To bridge this, researchers have begun fine-tuning open-weight LLMs specifically for quotation error detection. Huang et al. (2026) applied fine-tuned LLMs coupled with full-text analysis,

utilizing TokenSHAP to provide interpretability analysis. By highlighting exactly which tokens in the claim misalign with the reference text, this approach brings much-needed transparency to automated verification outputs. PubMed+ 1

### 4.3 Multi-Agent and Graph Frameworks

Perhaps the most significant recent development is the shift toward multi-agent, end-to-end autonomous frameworks. Systems like **BibAgent** integrate reasoning, retrieval, and adaptive evidence aggregation to track multi-hop miscitations. Addressing the persistent "paywall barrier" of full-text access, BibAgent deploys an "Evidence Committee" mechanism that infers the validity of a paywalled citation by analyzing the downstream citation consensus of other open-access papers. Evaluated on cross-disciplinary datasets (e.g., MisciteBench), such agentic frameworks have outperformed standard LLM baselines in both verification accuracy and scalability.

arXiv+ 2

## 5. Research Challenges and Limitations

Despite rapid algorithmic progress, automated detection of misinterpreted citations faces critical bottlenecks:

### **Scope Drift and Epistemic Overstatement:**

Unlike raw factual contradictions (e.g., stating a value increased when it decreased), citation misinterpretation often manifests as subtle scope drift. A reference finding that "Drug A reduced mortality in a cohort of 45 elderly male mice" might be cited as "Drug A is a proven life-extension therapy for mammals." Standard NLI models struggle with these boundary conditions, frequently misclassifying overstatements as valid support because the core semantic entities match.

### **The Full-Text Access (Paywall) Barrier:**

As highlighted by the developers of BibAgent, a massive

limitation in ACV is the inability to access full-text source data. When automated systems are restricted solely to freely available abstracts (as in the original SciFact design), they miss crucial context located in methods, supplementary files, or discussion sections, leading to false **IRRELEVANT** or **NOT ENOUGH INFO** classifications.

arXiv+ 1

### **Multimodal Evidence:**

Many scientific claims cite specific empirical results presented inside visual elements—such as survival curves, bar charts, or complex multi-variable regression tables. Text-only NLI engines cannot parse these structures, resulting in false positives when a claim is marked as unsupported simply because the evidence resides inside a figure.

## **6. Research Gaps and Future Directions**

To solidify the role of ACV in safeguarding academic integrity, future research must address the following gaps:

**Multimodal Scientific Claim Verification:** Integrating visual document understanding (VDU) into ACV pipelines will allow models to simultaneously analyze body text, embedded statistical tables, and vector charts to verify quantitative assertions.

**Real-Time Integration into Publishing Workflows:** Current models function primarily as retrospective auditing tools. Integrating automated citation verification natively into reference management software (e.g., Zotero) or manuscript submission portals (Editorial Manager) would allow authors to correct misinterpretations before publication.

**Standardized Benchmarks for Epistemic Modality:** While datasets like SciFact and the corpus from Sarol et al. (2024) are foundational, there is an urgent need for multi-disciplinary benchmarks that specifically isolate epistemic modality—testing a model's ability to differentiate between



"suggests," "indicates," and "proves" across different scientific domains. [Kaggle](#)

## 7. Conclusion

Misinterpreted but genuine citations represent a silent threat to scientific integrity. Because the underlying publications exist and possess valid metadata, these semantic misalignments easily bypass standard bibliographic checks, contaminating downstream research through self-perpetuating citation cascades.

Automated Claim Verification systems—driven by fine-tuned dense retrievers, natural language inference models, and emerging multi-agent frameworks like BibAgent—provide a technical mechanism to audit citation contexts at scale. While substantial challenges around full-text access, multimodal reasoning, and subtle overstatement detection remain, continuing advances in scientific NLP promise to transform automated citation auditing into a vital component of the modern academic publishing ecosystem.

[Semantic Scholar](#)+ 2

## Complete References

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The generated research paper is approximately **1,500 words** long, including the title, abstract, main body, and references.

are you warning me to verify ai generate citations?

Yes, absolutely. I highly recommend verifying every citation I generated before you use them in a real academic assignment.

While I followed your instructions to use genuine scholarly sources and avoid intentional fabrication—and the core papers by Agarwal, Greenberg, Sarol, and Wadden are real—as an AI, I am prone to occasional "hallucinations" or metadata mix-ups.

For example, an AI might:

- Pair a real author with a slightly incorrect publication year.
- Generate a DOI that looks structurally correct but leads to a dead link or a different paper.
- Confuse a preprint (like an arXiv ID) with the final published journal version.

Invent a recent date (like listing papers from 2026, which is an artifact of my generation process attempting to provide the most current research based on today's date).

### How to quickly verify them:

**Check the DOIs:** Paste the DOI link (e.g.,

`[https://doi.org/] (https://doi.org/) ...)`

directly into your browser. If it's a real paper, it will redirect to the publisher's page. **Search the Titles:** Copy and paste the title of the paper into Google Scholar or PubMed.

**Verify the Context:** Briefly skim the abstract of the real paper to ensure it actually supports the claim made in the text—which, ironically, is the exact problem of "misinterpreted genuine citations" discussed in the essay!

Always treat AI as an advanced drafting assistant, but act as your own peer-reviewer when it comes to bibliographic accuracy.